Project Executable Files

Date	02 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section documents all executable and deployable files for the OptiCrop project. All files are properly organized, named, and available on GitHub. The evaluator can access the live deployed application using the link below.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide	Yes
3	requirements.txt	Yes
4	SQLite Database (`database.db`)	Yes
5	Environment configuration (.env)	NA
6	Deployed application URL	NA (Local Deployment)
7	Executable/Application	Yes (Flask Web Application)
8	Model File (`crop_model.pkl`)	Yes
9	Testing Documents	Yes
10	Demo Video	Yes

Step 2: File / Folder Structure

```
OptiCrop/
│
├── 1. Brainstorming & Ideation/
│   ├── Brainstorming & Idea Prioritization.pdf
│   ├── Define Problem Statements.pdf
│   ├── Empathy Map.pdf
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│   └── ER Diagram.pdf
│
├── 2. Requirement Analysis/
│   ├── Customer Journey Map.pdf
│   ├── Data Flow Diagram.pdf
│   ├── Solution Requirements.pdf
│   └── Technology Stack.pdf
│
├── 3. Project Design Phase/
│   ├── Problem-Solution Fit.pdf
│   ├── Proposed Solution.pdf
│   └── Solution Architecture.pdf
│
├── 4. Project Planning Phase/
│   └── Project Planning.pdf
│
├── 5. Project Development Phase/
│   ├── OptiCrop.ipynb
│   ├── app.py
│   ├── train_model.py
│   ├── crop_model.pkl
│   ├── database.py
│   ├── database.db
│   ├── requirements.txt
│   ├── Crop_recommendation.csv
│   ├── templates/
│   │   ├── index.html
│   │   ├── login.html
│   │   ├── register.html
│   │   └── dashboard.html
│   ├── static/
│   │   ├── css/
│   │   ├── images/
│   │   └── js/
│   └── README.md
│
├── 6. Project Testing/
│   └── Performance Testing.pdf
│
├── 7. Project Documentation/
│   ├── Project Executable Files.pdf
│   └── Project Documentation.pdf
│
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│
├── 8. Project Demonstration/
│ ├── Communication.pdf
│ ├── Demonstration of Proposed Features.pdf
│ ├── Project Demo Planning.pdf
│ ├── Scalability & Future Plan.pdf
│ └── Team Involvement in Demonstration.pdf
│
└── README.md

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	**Not Deployed (Runs Locally)**
Login Credentials (Demo)	Register a new account using the **Register** page, then log in with your registered email and password.
Platform / Hosting Provider	Flask Development Server (Local Host)
Local Access URL	**http://127.0.0.1:5000**
Repository Link	https://github.com/geethikakonduru9/OptiCrop-Smart-Agricultural-Production-Optimization-Engine
Demo Video Link	https://drive.google.com/file/d/1pocAK2NtZauwpYRfDe05cirTQ1jD7PWj/view?usp=drive_link

Step 4: Run Instructions

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Option 1 — Run the Application Locally (Recommended):

Open the OptiCrop project folder in Visual Studio Code.
Install the required dependencies:
pip install -r requirements.txt

Run the Flask application:
python app.py

Open your browser and go to:
http://127.0.0.1:5000

2. **Option 2 — Run Locally:** 1. Clone the repository git clone https://github.com/geethikakonduru9/OptiCrop-Smart-Agricultural-Production-Optimization-Engine

3. Navigate to development folder cd "AI-ML-and-GEN-AI-Track-Project-Template/5.Project Development Phase"

4. Create and activate virtual environment python3 -m venv venv
source venv/bin/activate (Mac/Linux)

5. Install dependencies pip install flask scikit-learn pandas numpy matplotlib seaborn jupyter

6. Run Jupyter Notebook or vs code to train and save model jupyter notebook OptiCrop.ipynb → Run all cells → model.pkl will be generated

7. Run Flask application python app.py

8. Open browser: http://127.0.0.1:5000

Step 5: Known Issues / Limitations

- Local deployment only
- Accuracy depends on input data
- Supports crops available in the training dataset
- Future enhancement: weather API integration and cloud deployment